

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital VLSI Testing (01CT0629)	Aim: To model and detect stuck-at-0 and stuck-at-1 faults in 4-bit SISO (Serial In Serial Out) Register.	
Experiment No: 07	Date: 03/2026	Enrolment No: 92301733056

AIM: To model and detect stuck-at-0 and stuck-at-1 faults in 4-bit SISO (Serial In Serial Out) Register.

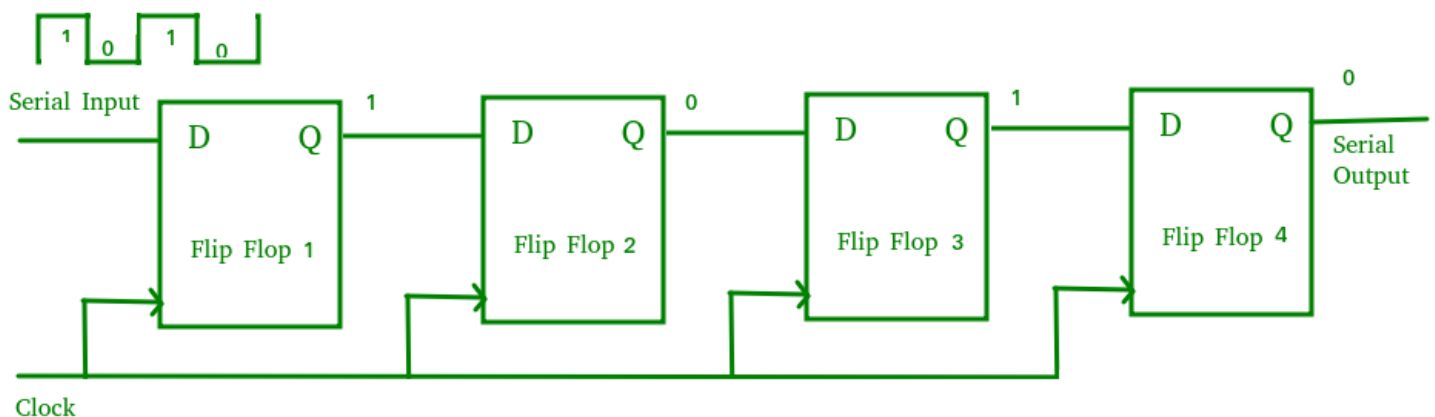
IDE: EDA Playground

THEORY:

- A SISO (Serial In Serial Out) shift register is a sequential circuit in which data is shifted bit-by-bit through flip-flops on each clock pulse. In a 4-bit SISO register, 4 flip-flops are connected in series, and data moves from one stage to the next.
- Fault Modeling:
 - Stuck-at-0 fault (SA0): Signal is permanently fixed at logic 0.
 - Stuck-at-1 fault (SA1): Signal is permanently fixed at logic 1.
 - These faults are commonly used in digital circuit testing to verify reliability.
- Fault Detection:
 - Apply test inputs
 - Observe output
 - Compare with expected output
 - If mismatch → fault detected

DESIGN AND IMPLEMENTATION:

A Serial-In Serial-Out shift register is a [sequential logic circuit](#) that allows data to be shifted in and out one bit at a time in a serial manner. It consists of a cascade of flip-flops connected in series, forming a chain. The input data is applied to the first flip-flop in the chain, and as the clock pulses, the data propagates through the flip-flops, ultimately appearing at the output. The logic circuit provided below demonstrates a serial-in serial-out (SISO) [shift register](#). It comprises four D flip-flops that are interconnected in a sequential manner. These [flip-flops](#) operate synchronously with one another, as they all receive the same clock signal.



4 Bit SISO Register

The synchronous nature of the flip-flops ensures that the shifting of data occurs in a coordinated manner. When the clock signal rises, the input data is sampled and stored in the first flip-flop. On

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subsequent clock pulses, the stored data propagates through the flip-flops, moving from one flip-flop to the next.

Each D flip-flop in the circuit has a Data (D) input, a Clock (CLK) input, and an output (Q). The D input represents the data to be loaded into the flip-flop, while the CLK input is connected to the common clock signal. The output (Q) of each flip-flop is connected to the D input of the next flip-flop, forming a cascade.

Working of SISO

A SISO is called simply because it has one input and one output. SISO stands for Single Input Single Output and is commonly used in relation to systems as well as signals. In plain terms, SISO means a system with only one input and one output.

The operating principle of SISO involves feeding the intended signal into the system where it gets processed in order to produce an output. Therefore, input signal refers to what goes into the system while output signal describes the result or response produced by the system depending on that particular input.

Signals at both the input and the output can be mathematically modeled by functions or equations in a SISO system. A system can be represented through a transfer function, describing how the system will modify the input signal to produce the output signal.

Different properties of the system, such as stability, frequency response, or time response, are taken into consideration to understand the behavior and performances of a system under different conditions.

Overall, a SISO system means that there is one input signal processed to obtain a corresponding output signal; therefore, it is central to the notion of systems and signals.

Truth Table for SISO

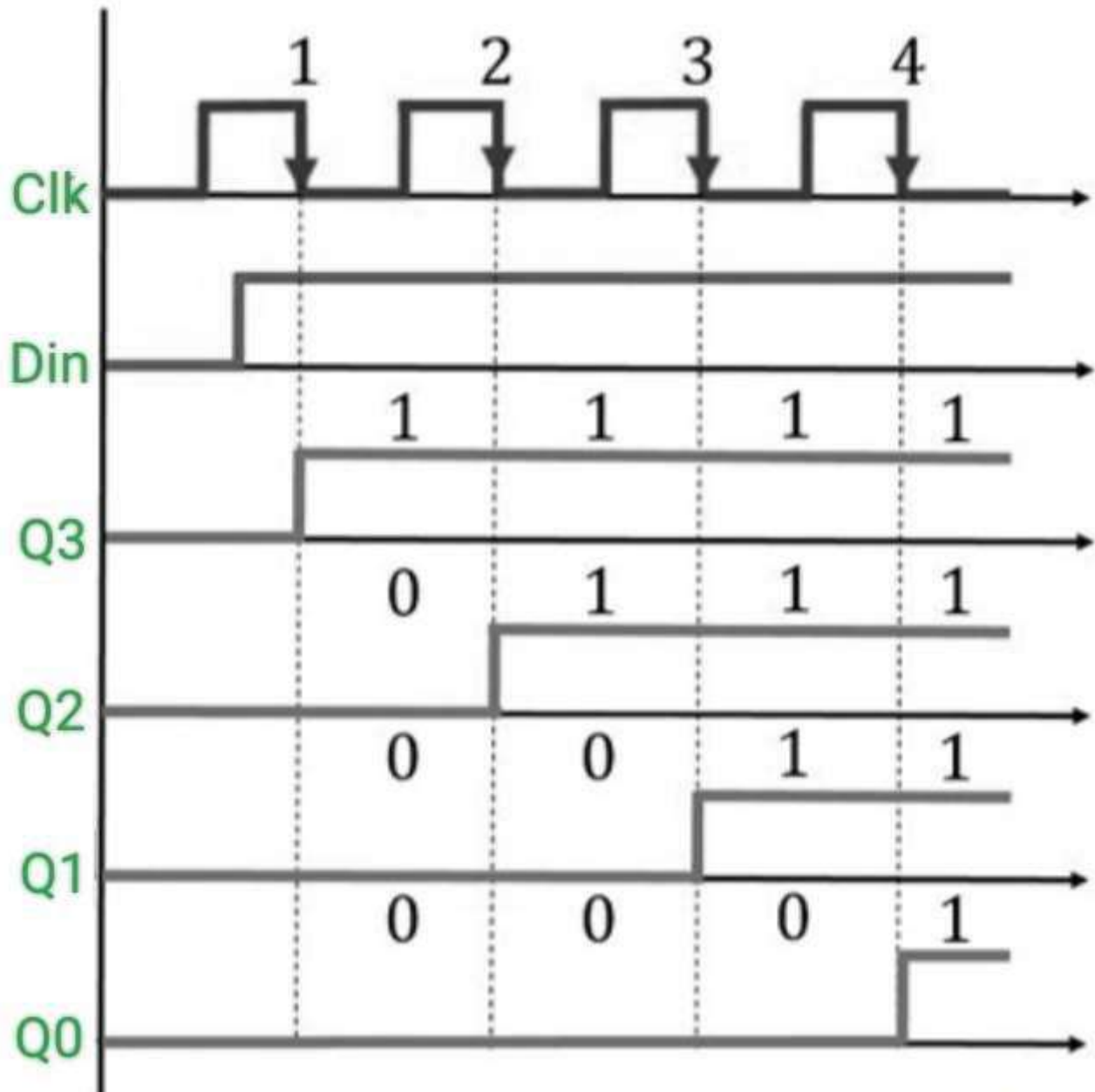
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CLK	Q3	Q2	Q1	Q0
Initial (Reset)	0	0	0	0
After 1st clock pulse	1	0	0	0
After 2nd clock pulse	1	1	0	0
After 3rd clock pulse	1	1	1	0
After 4th clock pulse	1	1	1	1

Truth Table

Waveform

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Waveform Representation

Functionality and Operation

The operation of a SISO shift register relies on two primary components: the flip-flops and the clock signal.

Flip-Flops: A flip-flop is a fundamental building block of sequential circuits. In the case of a SISO shift register, each [flip-flop](#) stores a single bit of data. The number of flip-flops determines the length or size of the shift register.

Clock Signal: The clock signal synchronizes the movement of data through the shift register. With each clock pulse, the data shifts from one flip-flop to the next. The clock signal ensures that the data propagates in a controlled and synchronized manner.

When the clock signal transitions from low to high (or high to low, depending on the specific implementation), the input data is sampled and stored in the first flip-flop. On subsequent clock pulses, the stored data moves through the chain of flip-flops. The output of the shift register is

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taken from the last flip-flop in the series.
Applications of SISO Shift Registers

VERILOG CODE:

A. 4-bit SISO Module

DESIGN CODE:

```

module siso_4bit (
    input clk,
    input sin,
    input sa0, // stuck-at-0 fault
    input sa1, // stuck-at-1 fault
    output sout
);

reg [3:0] q;

always @(posedge clk) begin
    if (sa0)
        q <= 4'b0000; // forced to 0
    else if (sa1)
        q <= 4'b1111; // forced to 1
    else
        q <= {q[2:0], sin}; // shifting operation
end

```

```

assign sout = q[3];

```

```

endmodule

```

B. Testbench

```

module tb_siso;

```

```

    reg clk;
    reg sin;
    reg sa0, sa1;
    wire sout;

```

```

    siso_4bit uut (
        .clk(clk),
        .sin(sin),
        .sa0(sa0),
        .sa1(sa1),
        .sout(sout)
    );

```

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```
// Clock generation
always #5 clk = ~clk;

initial begin
    $dumpfile("dump.vcd");
    $dumpvars(0, tb_siso);

    clk = 0;
    sin = 0;
    sa0 = 0;
    sa1 = 0;

    // Normal operation
    #10 sin = 1;
    #10 sin = 0;
    #10 sin = 1;

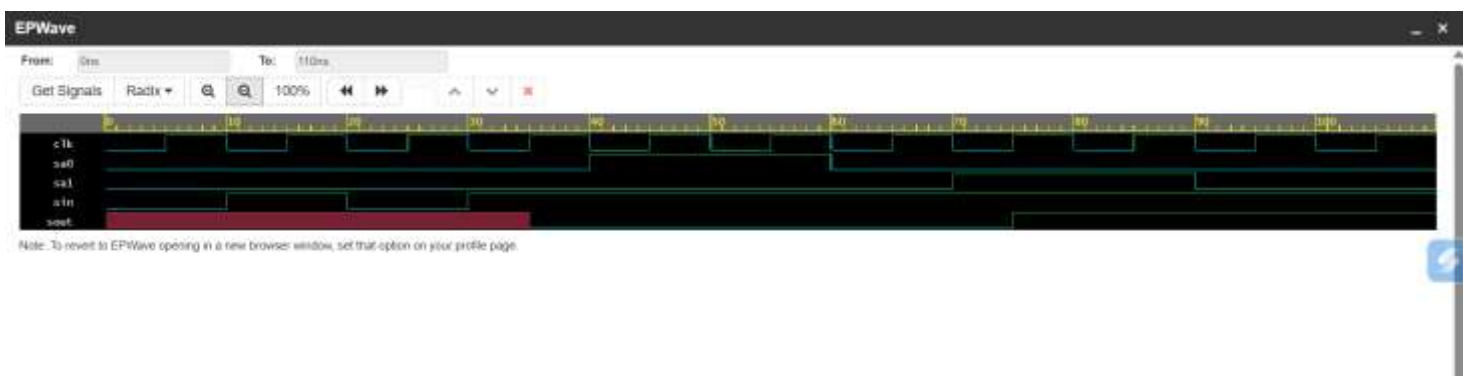
    // Stuck-at-0 fault
    #10 sa0 = 1;
    #20 sa0 = 0;

    // Stuck-at-1 fault
    #10 sa1 = 1;
    #20 sa1 = 0;

    #20 $finish;
end

endmodule
```

SIMULATION RESULTS:



CONCLUSION:

The 4-bit SISO shift register was successfully designed and simulated. The stuck-at-0 and stuck-at-1 faults were effectively modeled and detected. The simulation results confirmed that fault conditions alter the expected output, making fault detection possible. This experiment demonstrates the importance of fault modeling in digital system testing.